

HDOC 22

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Batch: 16

1. Algorithm

1. Start
2. Read two integers m and n .
3. Validate Inputs: If $m \leq 0$, $n \leq 0$, or $m > n$, display an error message and repeat step 2 until valid input is provided.
4. Display the menu with options 1 to 4.
5. Read user's choice.
6. Use a switch statement based on choice:
 - Case 1 (Tables side-by-side): Loop i from 1 to 10. In each iteration, print $m * i$ and $n * i$ in a formatted line.
 - Case 2 (Multiplication Grid): Print a top header row looping from 1 to n . Print a horizontal separator line. Loop i from 1 to m for the rows. Inside, print the row header i |, then loop j from 1 to n to print $i * j$.
 - Case 3 (Consecutive Integers): Loop i from m to n . Inside, print i ;; then loop j from 0 to 5, printing $i + j$.
 - Case 4 (Exit): Display termination message and break.
 - Default: Display "Invalid choice" error message.
7. Repeat steps 4-6 until choice == 4.
8. Stop

2. Test-Case Table

TC ID	Scenario Description	Inputs (m, n, choice)	Expected Output
TC-01	Invalid boundary input	m: 8, n: 3	Error: Ensure $m \leq n$ and both are positive integers.
TC-02	Tables side-by-side (Valid)	m: 3, n: 5, choice: 1	$3 \times 1 = 3$ $5 \times 1 = 5$... $3 \times 10 = 30$ $5 \times 10 = 50$
TC-03	Multiplication Grid (Valid)	m: 3, n: 5, choice: 2	Grid with 3 rows and 5 columns representing products 1x1 to 3x5.
TC-04	Consecutive Integers (Valid)	m: 3, n: 5, choice: 3	3: 3 4 5 6 7 8 4: 4 5 6 7 8 9 5: 5 6 7 8 9 10
TC-05	Invalid menu choice	choice: 7	Error: Invalid menu choice.
TC-06	Exit program	choice: 4	Program terminates correctly.

3. C Program Source Code

```
#include<stdio.h> int main() {
    int m,n, choice; int i, j;

    // Input m and n with validation do {
        printf("Enter two positive integers m and n (m <= n): "); scanf("%d %d", &m, &n);
        if (m <= 0 || n <= 0 || m > n) {
            printf("Error: Ensure m <= n and both are positive integers.\n");
        }
    } while (m <= 0 || n <= 0 || m > n);

    // Menu driven loop do {
        printf("\n1. Multiplication tables of m and n\n"); printf("2. Multiplication grid\n");
        printf("3. Consecutive integers\n"); printf("4. Exit\n");
        printf("Enter your choice: "); scanf("%d", &choice);

        switch (choice) { case 1:
            printf("Multiplication Tables of %d and %d (m=%d, n=%d)\n", m, n, m, n); for (i = 1; i <= 10; i++) {
                // %-2d and %-5d used for alignment
                printf("%d x %-2d = %-5d          %d x %-2d = %-5d\n", m, i, m * i, n, i, n * i);
            }
            break;

            case 2:
                printf("Multiplication Grid (m=%d, n=%d)\n", m, n); printf("      "); //
                Padding for top-left empty corner for (i = 1; i <= n; i++) {
                    printf("%4d", i);
                }
                printf("\n  --");
                for (i = 1; i <= n; i++) { printf("    ");
                }
                printf("\n");

                for (i = 1; i <= m; i++) { printf("%2d |", i);
                    for (j = 1; j <= n; j++) { printf("%4d", i *
                        j);
                    }
                    printf("\n");
                }
                break;
```

```

    case 3:
        printf("Consecutive Integers from %d to %d (m=%d, n=%d)\n", m, n, m, n);
        for (i = m; i <= n; i++) {
            printf("%d: ", i);
            for (j = 0; j <= 5; j++) {
                printf("%d ", i + j);
            }
            printf("\n");
        }
        break;

    case 4:
        printf("Exiting program...\n");
        break;

    default:
        printf("Error: Invalid menu choice.\n");
}
} while (choice != 4);

return 0;
}

```